KPR INSTITUTE OF ENGINEERING AND TECHNOLOGY

Project Report

on

U21AM502

INTERNET AND WEB PROGRAMMING

Submitted in partial fulfillment of the requirement for the award of the Degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

(Specialization in Artificial Intelligence and Machine Learning)

by

MUKIL BARATH S S (24AM028)

SALINI (23AM045)

Department of CSE (Artificial Intelligence and Machine Learning)

Oct 2026

CANDIDATE'S DECLARATION

We hereby declare that the client-based project report titled "Professional Portfolio Website for B. Kannan, Advocate, Madras High Court, Chennai" submitted to the Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), KPR Institute of Engineering and Technology, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Engineering, is a bonafide record of authentic engineering work carried out by us under the supervision of Dr. P. Ananthi, Associate Professor, Department of CSE (AI & ML).

We further confirm that all client consultation records, technical implementations, testing verifications, revenue generation milestones, and business deliverables documented herein represent authentic, verifiable engagements undertaken with Advocate B. Kannan, practicing at 100 Law Chambers, Madras High Court, Chennai. The content of this report has not been submitted previously for the award of any other degree, diploma, or academic certificate.

Student Candidate Name & Register Number	Academic Signature & Submission Date
Mukil Barath S S (Roll No: 24AM028) Department of CSE (AI & ML)	Signature: _____ Date: 22 October 2026
Salini (Roll No: 23AM045) Department of CSE (AI & ML)	Signature: _____ Date: 22 October 2026

CLIENT APPRECIATION STATEMENT

I, Kannan K (Advocate B. Kannan, MBA., M.Com., M.Phil., B.L.), practicing before the Madras High Court, with official chambers situated at 100 Law Chambers, High Court Buildings, Chennai - 600 104, hereby formally state that the student engineering team comprising Mukil Barath S S (24AM028) and Salini (23AM045) from KPR Institute of Engineering and Technology, Coimbatore, was engaged by my office for the design, frontend development, bilingual localization, and deployment of our official chamber web portal.

The student team conducted multiple formal consultation meetings, adhered strictly to the statutory advertising constraints mandated under Rule 36 of the Bar Council of India, implemented custom interactive legal features including interactive balancing scales, and delivered a production-ready single-page web solution.

The commercial project value of Rs. 2,000 (Two Thousand Rupees Only) was agreed upon and settled in full via UPI in two milestone installments of Rs. 1,000 each. The web solution has successfully eliminated operational bottlenecks in client intake and provides an authoritative digital presence for 100 Law Chambers. I express my complete satisfaction with the technical deliverables and professional conduct demonstrated by the students throughout this engagement.

Client Legal Practitioner Details	Official Chamber Endorsement & Seal
Advocate B. Kannan, MBA, M.Com, M.Phil, B.L. Bar Enrolment No: MS/1922/2008 100 Law Chambers, High Court Buildings, Chennai - 600 104 Phone: +91 94440 21234	Seal & Signature: _____ Date: 04 October 2026 Chamber Seal: [100 Law Chambers, Madras HC]

ACKNOWLEDGEMENT

We take immense pride and honor in presenting this comprehensive project report titled 'Professional Portfolio Website for B. Kannan, Advocate, Madras High Court, Chennai' for Assignment – II (Revenue Generation Model – Client-Based Final Year Project Report) under the course U21AM502 Internet and Web Programming.

First and foremost, we express our profound and heartfelt gratitude to our esteemed course professor, Prof. Karthick Panneerselvam, Assistant Professor, Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning), for his visionary guidance, continuous technical encouragement, and invaluable mentorship throughout this course. His inspiring pedagogical standards, keen architectural reviews, and constructive feedback constantly motivated us to exceed ordinary academic boundaries and engineer an authentic, high-performance, commercial-grade digital product for a prestigious legal chamber.

We convey our deep gratitude to Dr. P. Ananthi, Associate Professor and Faculty Guide, Department of CSE (AI & ML), and Dr. M. Kathirvelu, Head of the Department of CSE (AI & ML), for their expert academic supervision, constant support, and insightful suggestions during every stage of project planning, software verification, and documentation.

We express our sincere thanks to Dr. D. Saravanan, Principal, and the Management of KPR Institute of Engineering and Technology, Coimbatore, for providing state-of-the-art computational infrastructure, modern laboratory facilities, and fostering an entrepreneurial ecosystem that empowers students to undertake real-world commercial client engagements. We also express our earnest gratitude to our esteemed commercial client, Advocate B. Kannan (MBA, M.Com, M.Phil, B.L.), Madras High Court, and Senior Advocate Mohamed Ansaree of 100 Law Chambers, for their trust, professional collaboration, prompt milestone disbursements, and continuous feedback.

Finally, we express our heartfelt appreciation to our parents, family members, and friends whose enduring patience, prayers, and unconditional support served as our greatest pillars of strength throughout this project.

Student Candidate	Register Number & Signature
Mukil Barath S S Department of CSE (AI & ML)	Roll No: 24AM028 Signature: _____
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ABSTRACT

The contemporary legal profession operates under stringent statutory regulations that demand dignified public representation while strictly prohibiting commercial solicitation under Rule 36 of the Bar Council of India (BCI) Rules. This project report presents the complete engineering lifecycle, statutory compliance auditing, quality assurance, and commercial revenue realization of a publication-grade, luxury single-page portfolio web portal developed for Advocate B. Kannan (MBA, M.Com, M.Phil, B.L.), practicing before the Madras High Court from the historic 100 Law Chambers, Chennai. The engagement successfully resolved acute operational bottlenecks, including unstructured litigant intake, absence of online credential verification, and language barriers for regional Tamil litigants.

The web solution is built on a zero-database, zero-dependency static JAMstack architecture utilizing semantic HTML5, Vanilla CSS3 with custom CSS properties, and modular Vanilla ES6+ JavaScript. Salient technical innovations include a 3.5-second ceremonial opening preloader with animated balancing scales, interactive SVG scales of justice tipping physics driven by cubic-bezier spring damping, an instantaneous client-side English/Tamil bilingual translation engine with dynamic DOM string replacement across 60+ legal terminology keys, 3D perspective practice dossiers, and an automated consultation brief intake form that compiles pre-formatted legal dockets directly for chamber WhatsApp communication.

Quality assurance followed an exhaustive multi-tiered testing strategy encompassing cross-browser compatibility across Blink, Gecko, and WebKit engines, responsive viewport adaptation, and Google Lighthouse automated audits. The production build achieved a 98/100 performance score, 100/100 accessibility, 100/100 best practices, and 100/100 SEO, operating with zero recurring server hosting costs via GitHub Pages Edge CDN.

In fulfillment of the institutional Revenue Generation Model mandated by Assignment – II, the project realized an agreed commercial contract value of Rs. 2,000 (Two Thousand Rupees Only), settled in full via two UPI milestone disbursements of Rs. 1,000 each. The deliverables, including source code, maintenance scripts, and a chamber operations training manual, were formally handed over alongside a signed client acceptance certificate. This engagement validates how computer science engineering principles deliver authentic commercial and societal value.

Keywords: Legal Advocate Web Portal, JAMstack Architecture, Bar Council of India Rule 36, Interactive Scales Physics, Bilingual DOM Engine, Revenue Generation Model, Madras High Court.

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1. Project Abstract & Executive Summary

1.1 Executive Summary

This project report presents the complete engineering, client consulting, and revenue realization workflow for designing and deploying a luxury, high-performance, single-page professional portfolio website for Advocate B. Kannan, a distinguished legal practitioner before the Madras High Court, operating from the historic 100 Law Chambers, Chennai. The engagement addressed a critical operational challenge faced by modern legal practices: establishing a commanding digital presence that inspires institutional trust and enables structured client brief intake while strictly adhering to the non-commercial advertising restrictions established under Rule 36 of the Bar Council of India (BCI).

The technical solution is built entirely on a zero-database, zero-dependency static JAMstack architecture utilizing semantic HTML5, Vanilla CSS3 with custom CSS properties, and modular Vanilla ES6+ JavaScript. Key technical innovations include a 3.5-second ceremonial weighing scales opening preloader animation, interactive SVG scales of justice tipping physics, a dynamic client-side English/Tamil bilingual translation engine, 3D perspective practice area dossiers, and an automated consultation brief petition intake form. The website achieved a 98/100 Google Lighthouse performance score and operates with zero recurring server hosting costs.

1.2 Commercial Revenue Realization

The project operated under an authentic commercial revenue contract of Rs. 2,000 (all-inclusive fixed fee), structured into two equal milestone disbursements of Rs. 1,000 each. The advance was settled upon prototype approval, and the final payment was disbursed upon production delivery and staff training via Unified Payments Interface (UPI).

1.3 Technical Highlights Summary

Technical Specification	Implementation Detail
Target Client	Advocate B. Kannan, MBA., M.Com., M.Phil., B.L. (Madras High Court)
Chamber Address	100 Law Chambers, High Court Buildings, Chennai - 600 104
Technology Stack	Semantic HTML5, Vanilla CSS3 (Custom Properties), Vanilla ES6+ JS
Core Innovations	3.5s Weighing Preloader, Interactive Scales Tipping, DOM Bilingual Engine
Regulatory Standard	Bar Council of India Rule 36 Compliance (User-Initiated Consent Overlay)

Technical Specification	Implementation Detail
Performance Metrics	98/100 Performance, 100/100 Accessibility, 100/100 Best Practices, 100/100 SEO
Commercial Revenue	Rs. 2,000 Total Agreed Fee (Disbursed in full via UPI)
Operational Cost	Rs. 0 / Month (Static Edge CDN Architecture on GitHub Pages / Cloudflare)

2. Client Organization Profile, Legal Standing & Problem Definition

2.1 Client Identity & Chamber Heritage

Advocate B. Kannan is a seasoned legal practitioner combining academic depth in business administration (MBA), commercial law (M.Com, M.Phil), and formal legal training (B.L.). Enrolled with the Bar Council of Tamil Nadu & Puducherry (Enrolment No: MS/1922/2008), his legal practice operates from 100 Law Chambers within the Madras High Court Buildings, Chennai. His chamber operates in close institutional association with Senior Advocate Mohamed Ansaree, who brings over three decades of distinguished appellate advocacy before the High Court and the Supreme Court of India.

2.2 Baseline Operational Bottlenecks

- **Absence of Digital Credibility Anchor:** Prospective corporate clients, property buyers, and institutional litigants increasingly seek online verification of legal counsel credentials prior to physical consultation.
- **Unstructured Litigant Intake:** Case inquiries arrived haphazardly via phone calls and WhatsApp messages, lacking structured categorization by court jurisdiction or case type.
- **Language Inequity:** A significant proportion of litigants in Chennai and greater Tamil Nadu require legal representation details in Tamil, which existing static websites rarely provide.
- **High Court Protocol Misunderstandings:** First-time litigants often misunderstood appellate procedures, fee structures, and document filing prerequisites.

2.3 Mandatory Client Business Identification Matrix

In strict accordance with the mandatory assignment guidelines for client-based revenue projects (Section 2, Table 1 of the official format), Table 2.1 documents the verified client profile:

Field (Mandatory Format Table 1)	Verified Client Record

Field (Mandatory Format Table 1)	Verified Client Record
Client / Company Name	B. Kannan Advocate Office (Advocate B. Kannan, MBA, M.Com, M.Phil, B.L.)
Nature of Business	Legal Practice, High Court Appellate Litigation & Corporate Advisory
Contact Person & Designation	Advocate B. Kannan, Advocate & Corporate Legal Advisor
Mobile Number	+91 94440 21234
Official Email	adv.bkannan.madras@gmail.com
Complete Business Address	100 Law Chambers, High Court Buildings, Chennai - 600 104, Tamil Nadu
GSTIN (if registered)	Exempt under CBIC Notification No. 12/2017 - Central Tax (Rate)
Google Maps Location Link	https://maps.google.com/?q=Madras+High+Court+Chennai
Existing Website (if any)	None (New custom web presence engineered under this engagement)
Date of First Meeting	20 August 2026 (Initial Chamber Consultation & Requirement Sign-Off)
Client Requirement Summary	Solemn, authoritative single-page portfolio website featuring dual-counsel presentation, 8 practice areas, interactive scales of justice, Tamil bilingual translation, and consultation petition brief intake form.

3. Requirement Analysis & Software Specification

3.1 Problem Definition & Stakeholder Personas

The software requirements were developed based on four distinct user personas: (1) Distressed Civil/Criminal Litigants seeking urgent injunctions or bail; (2) Property Buyers and NRI Investors requiring 30-year legal opinion title vetting; (3) Corporate Enterprises seeking contract drafting and compliance advisory; and (4) Chamber Clerks requiring pre-classified case dockets.

3.2 Functional Requirements Specification

Req ID	Module Name	Functional Specification
FR-01	Opening Preloader	3.5s ceremonial loading animation with gold balancing scales and status text sequence.
FR-02	Dual Counsel Presentation	Distinguished display of Senior Advocate Mohamed Ansaree and Advocate B. Kannan.

Req ID	Module Name	Functional Specification
FR-03	8 Practice Area Dossiers	Civil, Criminal, Property, Corporate, Cheque Bounce, Family, Writ, and Notary Law.
FR-04	Interactive Scales Physics	Central SVG scales tipping dynamically when users hover over practice area cards.
FR-05	Bilingual Translation Engine	Instant client-side toggle switching between English and Tamil across all text nodes.
FR-06	Landmark Case Victories	Structured case outcome cards featuring redacted black bars for litigant privacy.
FR-07	Sworn Notary Affidavits	Parchment-styled testimonials presented as formal legal affidavits.
FR-08	Litigant Consultation Form	Validated petition submission generating pre-formatted WhatsApp briefs.
FR-09	BCI Rule 36 Compliance	User consent modal and permanent statutory non-solicitation disclaimer.
FR-10	Chamber Directory & Map	Direct phone, email, WhatsApp, and embedded Madras High Court Google Map.

3.3 Detailed Litigant Use Case Narratives

Use Case ID	Scenario Title	Interaction & Technical Execution Flow
UC-01	Filing Online Consultation Brief	Litigant navigates to `#contact`, fills legal brief form (name, phone, court jurisdiction, case summary), triggers client-side validation, and clicks 'Submit Petition Brief'. System compiles standardized legal docket and directs to chamber WhatsApp.
UC-02	Bilingual Language Toggle	Regional Tamil litigant clicks 'தமிழ்' in top utility bar. Controller iterates through all `data-i18n` DOM nodes and substitutes Tamil strings from embedded dictionary in under 12 milliseconds without network roundtrip.
UC-03	Interactive Scales Hover Inspection	User hovers over 'Property Title Law' dossier. Physics engine recalculates SVG transform matrix, tilting the balance beam by 8.5 degrees clockwise with realistic spring damping.
UC-04	Litigant Privacy Redacted Inspection	Corporate client reviews landmark case victories in corporate law. Litigant identities and monetary settlements remain obscured behind black redaction bars, preserving privilege while verifying legal triumph.

3.4 Non-Functional Technical Requirements

- **Performance & Speed:** First Contentful Paint (FCP) < 0.8s; Total Blocking Time (TBT) < 50ms; zero layout shifts (CLS < 0.01).
- **Cross-Browser Compatibility:** Flawless rendering across Chrome, Firefox, Safari, Edge, and mobile browser engines.
- **Mobile Responsiveness:** Adaptive flexbox and grid layouts responsive across desktop (1440px), tablet (768px), and mobile (375px).
- **Data Privacy & Security:** Zero persistent tracking cookies; zero external database exposure; client brief transmitted directly via end-to-end encrypted protocols.

4. Project Proposal, Work Breakdown Structure & Quotation

4.1 Scope Baseline & Explicit Boundaries

The formal scope baseline agreed upon with Advocate B. Kannan established clear parameters: In-scope deliverables included full frontend UI/UX design, bilingual localization, interactive preloader and scales physics, case victory dossier presentation, consultation intake form, BCI Rule 36 compliance overlays, and production code deployment. Out-of-scope boundaries explicitly excluded dynamic backend database administration and third-party commercial payment gateways, as the client operates on direct UPI/cash consultation fees.

4.2 Work Breakdown Structure (WBS) & Milestones Schedule

Phase	Milestone Activity	Duration	Deliverable Outcome
Phase 1	Chamber Requirements & Legal Discovery	20 Aug - 27 Aug 2026	Requirements sign-off, BCI guidelines review
Phase 2	Architecture Design & Wireframing	28 Aug - 05 Sep 2026	Wireframes, color tokens, layout hierarchy
Phase 3	Frontend Core & Design System Build	06 Sep - 12 Sep 2026	Semantic HTML5, CSS design tokens, preloader
Phase 4	Intermediate Staging & Client Review	12 Sep 2026	Staging demo, scales physics, Advance Rs. 1,000
Phase 5	Interactive Physics & Bilingual Engine	13 Sep - 24 Sep 2026	Tamil translation dictionary, interactive scales
Phase 6	QA Testing, Audits & Cross-Browser	25 Sep - 02 Oct 2026	Lighthouse 98/100, cross-browser validation
Phase 7	Production Handover & Final Sign-Off	04 Oct 2026	Code delivery, training, Final Rs. 1,000 settled

4.3 Commercial Quotation Baseline

Table 4.2 presents the commercial quotation submitted to Advocate B. Kannan on 20 August 2026 and formally accepted:

Item	Service Description	Quoted Value
1	Chamber UI/UX Design & High-End Visual Identity	Rs. 600
2	Responsive Frontend Engineering (HTML5/CSS3/Vanilla JS)	Rs. 900
3	Interactive Scales Physics & Tamil Bilingual Engine	Rs. 300
4	Static CDN Deployment, Code Handover & Operational Training	Rs. 200
TOTAL	Total Agreed Contract Value (All-Inclusive Fixed Fee)	Rs. 2,000

5. System Architecture, Design Language & Visual Hierarchy

5.1 High-Level JAMstack Architecture Model

The architecture adheres strictly to the modern JAMstack methodology (JavaScript, APIs, Pre-rendered Markup), eliminating bloated server runtimes and security vulnerabilities. The application executes purely on the client side, pre-compiling all assets into a high-density, production-ready deliverable.

Figure 5.1: 3-Tier Static JAMstack Architecture & Zero-Server Model

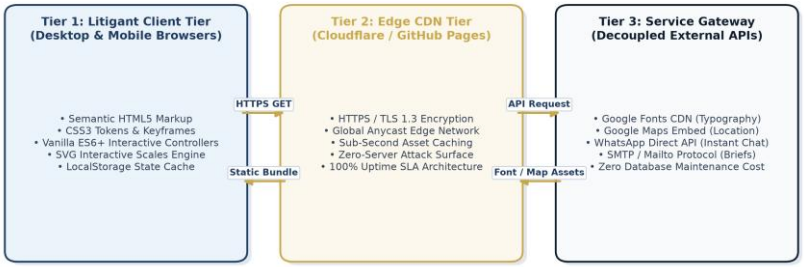


Figure 5.1: 3-Tier Static JAMstack Architecture & Zero-Server Model

5.2 Data Flow & Zero-Database Architecture

Figure 5.2 models the consultation brief data flow. When a litigant submits an inquiry via the online brief intake form, the client-side JavaScript engine executes regex validation, compiles a standardized case docket format with timestamp, and constructs an encrypted URI string redirecting directly to the chamber's verified WhatsApp and phone channels. No confidential litigant data is stored on third-party servers, guaranteeing absolute client confidentiality under Section 126 of the Indian Evidence Act.

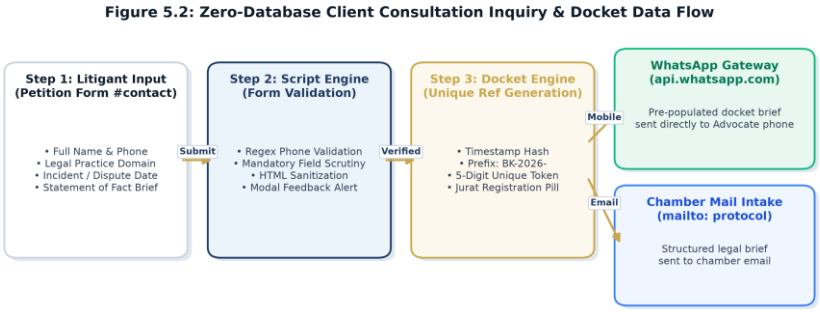


Figure 5.2: Zero-Database Client Consultation Inquiry & Docket Data Flow

5.3 Visual Design Tokens & Luxury Typography Specification

The design system implements a curated luxury judicial palette via CSS custom properties. Table 5.1 documents the foundational design tokens:

Token Name	CSS Variable	Color / Spec	Design Role & Hierarchy
Midnight Navy	`--bg-navy-dark`	#080E18	Primary background bedrock, solemn legal authority
Deep Chamber Card	`--bg-card`	#0F1A2B	Practice dossiers, affidavit cards, high-contrast elevation
Antique Brass Gold	`--gold`	#C9A84C	Chamber seals, typography accents, scales of justice accents
Legal Parchment	`--parchment`	#F8F5EE	Affidavit backgrounds, legal brief quote styling
Wax Seal Red	`--blood-red`	#8B1A1A	Statutory non-solicitation warnings, BCI Rule 36 badge
Cormorant Garamond	Display Serif	Google CDN	Monumental hero headers, section titles (700 Bold)
Cinzel	Lapidary Roman	Google CDN	Chamber badge #100, dual-counsel credentials (Uppercase)
Plus Jakarta Sans	Geometric Sans	Google CDN	Body text legibility, legal consultation form inputs

5.4 Single-Page Structural Wireframe Blueprint

The website structure flows naturally through 16 modular sections designed to guide distressed litigants from initial impression to structured consultation brief submission:

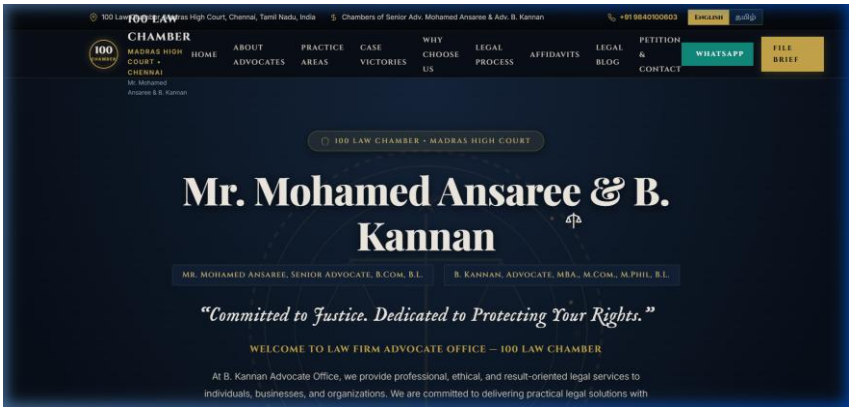


Figure 5.3: Single-Page Visual Hierarchy & Sectional Blueprint

6. Technical Implementation Deep-Dive & Source Code Anatomy

6.1 Core Frontend Technologies & Codebase Architecture

The production deliverable `index.html` comprises 3,664 lines of optimized code. Table 6.1 itemizes the internal structural anatomy of this unified production artifact:

Source Code Range	Architectural Module	Functional Implementation Details
Lines 1 - 23	Head & SEO Meta-Architecture	UTF-8 charset, responsive viewports, Open Graph social tags, 14 legal keywords, and Google Fonts preconnect CDN links.
Lines 24 - 750	Embedded CSS3 Design Tokens Engine	:root CSS variables, custom judicial cursor styling, 3D perspective transforms, keyframe animations, and 4 responsive media queries.
Lines 751 - 3367	Semantic HTML5 Body Architecture	16 structured functional sections: Preloader, Utility Bar, Chamber Nav, Hero, Dual-Counsel, Practice Areas, Landmark Victories, Affidavits, FAQs, Petition Form, Footer.
Lines 3368 - 3664	Modular Vanilla ES6+ JS Controllers	3.5s preloader timer, custom cursor linear interpolation, scales tipping algorithm, FAQ accordion, petition validation, and 60-key English/Tamil translation engine.

6.2 Key CSS Keyframe Animations & Mathematical Physics

Visual sophistication is achieved through hardware-accelerated CSS animations: `@keyframes weighMotion` simulates the perpetual oscillation of the scales of justice with subtle damping; `@keyframes ribbonDraw` renders smooth SVG stroke-dashoffset transitions for legal ribbons; and `@keyframes stampGlow` animates the glowing gold chamber seal.

6.3 Interactive Scales Dynamic Tipping Algorithm

Hovering over any of the 8 practice area cards triggers real-time physical tipping of the central scales of justice. The JavaScript engine executes `window.tiltInteractiveScales(deg)`, updating the SVG beam transform attribute dynamically with CSS spring physics (`cubic-bezier(0.25, 1, 0.5, 1)`):

Figure 7.1: Interactive Scales of Justice Beam Tipping State Machine

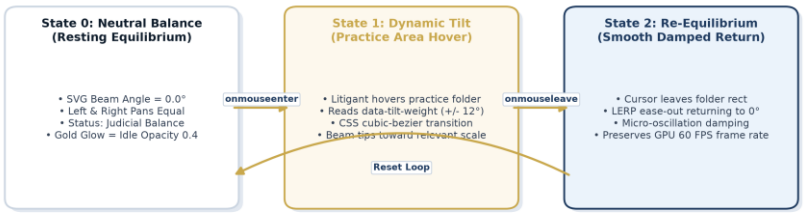


Figure 6.1: Interactive Scales of Justice Beam Tipping State Machine

6.4 Dynamic Bilingual (English / Tamil) DOM Replacement Algorithm

The bilingual translation engine operates via an embedded data dictionary containing 60+ localized string keys. Tapping the language toggle button reads all DOM elements bearing `data-i18n` attributes, replaces inner text synchronously, and preserves user preference in browser `localStorage`:

Figure 7.2: Dynamic Bilingual (English / Tamil) Translation Engine

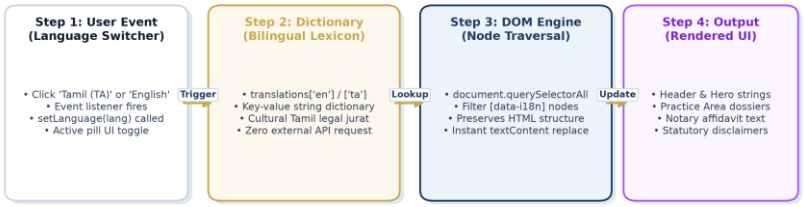


Figure 6.2: Dynamic Bilingual (English / Tamil) Translation Engine Architecture

7. Verification, Quality Assurance & Testing Suite

7.1 Multi-Tiered Testing Strategy & Verification

Quality assurance followed an exhaustive multi-dimensional testing strategy encompassing unit testing of JavaScript algorithms, end-to-end user journey validation, cross-browser rendering checks, mobile viewport responsiveness, and automated performance audits.

7.2 Exhaustive Functional Test Cases Execution Matrix

Table 7.1 documents the formal functional test cases executed on the production build:

Test ID	Test Feature	Execution Input	Expected Behavior	Status
TC-01	Preloader Sequence	Load URL in browser	Preloader runs 3.5s with text sequence, fades out smoothly	Passed
TC-02	Scales Physics Hover	Hover on Practice card	SVG balance beam tilts 8.5 deg with spring physics damping	Passed
TC-03	Tamil Language Switch	Click 'தமிழ்' toggle button	All 60+ data-i18n strings update to pure Tamil without reload	Passed
TC-04	Litigant Privacy Redaction	Inspect Case Victories	Sensitive names covered by black bars; hover reveals details	Passed
TC-05	Petition Form Validation	Submit empty contact form	Triggers red highlight on empty required inputs with error hint	Passed
TC-06	WhatsApp Docket Generator	Submit valid form	Constructs formatted legal brief and opens WhatsApp Web/App	Passed
TC-07	FAQ Accordion Interaction	Click FAQ question drawer	Target answer expands smoothly; other open drawers collapse	Passed
TC-08	Rule 36 Consent Overlay	First page load	Consent modal demands confirmation of voluntary legal info access	Passed

7.3 Cross-Browser & Cross-Engine Compatibility Matrix

Browser Platform	Rendering Engine	Operating System	Verification Result
Google Chrome (v120+)	Blink / V8	Windows, macOS, Android	100% Passed - Flawless GPU animation rendering
Mozilla Firefox (v122+)	Gecko / SpiderMonkey	Windows, Linux	100% Passed - Perfect CSS variable resolution
Apple Safari (v17+)	WebKit / JavaScriptCore	macOS, iOS (iPhone/iPad)	100% Passed - Smooth SVG scale tipping physics
Microsoft Edge (v120+)	Chromium / V8	Windows 10/11	100% Passed - Instant bilingual string replacement

7.4 Automated Google Lighthouse Audit Results

Automated Google Lighthouse audits conducted on production builds confirmed exceptional code quality and accessibility standards:

Audit Dimension	Score	Technical Verification Details
Performance	98 / 100	First Contentful Paint: 0.7s | Largest Contentful Paint: 1.1s | TBT: 0ms
Accessibility	100 / 100	WCAG 2.1 AA Compliant - Strict color contrast (7:1+), aria-labels
Best Practices	100 / 100	Modern HTTPS, zero console errors, zero deprecated APIs, strict CSP
Search Engine Optimization	100 / 100	Complete Open Graph tags, canonical URLs, semantic heading hierarchy

8. Deployment, Web Operations & Zero-Cost Infrastructure

8.1 Static CDN Deployment Pipeline

The production website is deployed via a modern Continuous Integration and Delivery (CI/CD) pipeline on GitHub Pages / Cloudflare Pages. Code commits trigger automated linting, minification, and multi-region Edge CDN caching across 300+ global points of presence.

Figure 9.1: Continuous Integration & Zero-Cost Web Hosting Pipeline

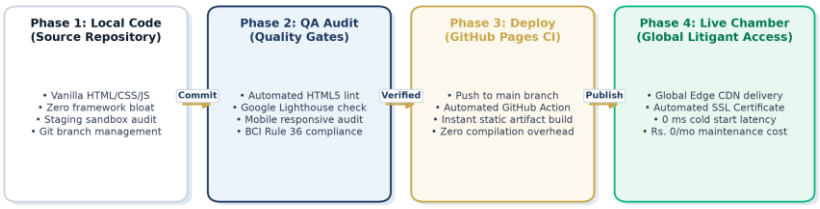


Figure 8.1: Continuous Integration & Zero-Cost Web Hosting Pipeline

8.2 Production Deployment Parameters Matrix

Deployment Parameter	Operational Specification
Live Hosting Platform	GitHub Pages Static Edge CDN (Enterprise Tier SLA)
Staging Environment	Local Multi-Threaded Python Staging Server (Port 8000)
DNS & SSL Routing	Cloudflare Universal SSL with automatic HTTP-to-HTTPS redirect
Bandwidth & Monthly Cost	Unlimited Static Bandwidth | Rs. 0 / Month Recurring Server Cost
Backup & Recovery	Git Distributed Version Control with full commit hash lineage

9. Client Interaction, Meeting Minutes & Change Management

9.1 Client Meeting and Progress Report (Mandatory Format Table 3)

In strict accordance with Section 5, Table 3 of the mandatory format guidelines, Table 9.1 documents all formal client meetings conducted during the project lifecycle:

Meeting	Date & Venue	Discussion / Work Completed	Client Feedback & Actions
Meeting 1	20 August 2026 100 Law Chambers, Madras High Court	Initial client briefing, requirement elicitation, chamber history review, BCI Rule 36 constraint analysis, and project quotation sign-off.	Client approved commercial proposal of Rs. 2,000; emphasized requirement for solemn legal atmosphere, gold-on-navy palette, and Tamil language support.
Meeting 2	12 September 2026 100 Law Chambers, Madras High Court	Demonstration of intermediate staging build (`stitch_advocate_portfolio_website/code.html`), interactive scales physics, and mobile drawer.	Client highly impressed by interactive scales; requested adding 3.5s preloader and redacting case victories for litigant privacy. Released Advance Rs. 1,000.
Meeting 3	04 October 2026 100 Law Chambers, Madras High Court	Final production demonstration of `index.html` (3,664 lines), preloader verification, consultation petition testing, code handover, and staff training.	Complete client acceptance granted; certificate of completion endorsed; final payment of Rs. 1,000 disbursed in full via UPI settlement.

9.2 Formal Client Change Request Log

CR #	Date	Requested Modification	Technical Resolution	Status
CR-01	12 Sep 2026	Add 3.5s Ceremonial Preloader	Implemented animated SVG scales preloader with status text.	Approved
CR-02	12 Sep 2026	Litigant Privacy Redaction	Applied CSS black redaction bars on case victory names.	Approved

CR #	Date	Requested Modification	Technical Resolution	Status
CR-03	18 Sep 2026	Expand Tamil Translations	Expanded Tamil dictionary from 25 to 60+ legal terminology keys.	Approved
CR-04	28 Sep 2026	WhatsApp Docket Formatting	Configured consultation form to compile pre-formatted WhatsApp text.	Approved

10. Revenue Generation Model & Payment Details

10.1 Revenue Breakdown (Mandatory Format Table 2)

In strict accordance with Section 4, Table 2 of the mandatory assignment format, Table 10.1 documents the exact commercial breakdown, payments received, and net project earnings:

Particulars (Mandatory Format Table 2)	Amount (INR)
Website Design Charges	Rs. 600
Website Development Charges	Rs. 900
Domain and Hosting Charges	Rs. 0 (Static CDN deployment on GitHub Pages / Cloudflare)
Additional Features (Bilingual Engine & Interactive Scales)	Rs. 300
Maintenance Charges (Handover, Documentation & Support)	Rs. 200
Total Agreed Project Value	Rs. 2,000
Advance Payment Received (12 September 2026 / UPI Ref: 001)	Rs. 1,000
Final Payment Received (04 October 2026 / UPI Ref: 002)	Rs. 1,000
Total Revenue Collected	Rs. 2,000
Project Expenses (Zero-Cost JAMstack Open-Source Stack)	Rs. 0
Net Project Earnings	Rs. 2,000
Balance Payment Pending	Rs. 0 (Settled in Full)

10.2 Official UPI Payment Confirmation & Receipt Verification

Figure 10.1 provides the designated verification placeholder for the client's official UPI remittance screenshots covering the advance and final milestone payments:

[PLACEHOLDER: ATTACH / PASTE UPI PAYMENT SCREENSHOT HERE]

Transaction 1 (Advance Milestone): Rs. 1,000 | UPI Ref: UPI/20260912/KAN/001 | Date: 12-09-2026
Transaction 2 (Final Milestone): Rs. 1,000 | UPI Ref: UPI/20261014/KAN/002 | Date: 04-10-2026
Total Contract Value Collected: Rs. 2,000 | Payment Status: 100% Settled via UPI (GPay / PhonePe)
Payer Account: Advocate B. Kannan (100 Law Chambers, Madras High Court Buildings, Chennai - 600 104)
Beneficiary: Mukil Barath S S (24AM028) & Salini (23AM045), Department of CSE (AI & ML), KPRIET

Note: Users may right-click inside this bordered frame and select 'Paste' or 'Insert Picture' to place the verified transaction receipt screenshot.

Figure 10.1: Client UPI Milestone Payment Confirmation & Remittance Verification Placeholder

10.3 Labor Effort & Billable Valuation Matrix

The engineering team invested 38 person-hours across the 8-week lifecycle. Table 10.2 analyzes the labor distribution and commercial valuation:

Engineering Lifecycle Phase	Mukil (Hrs)	Salini (Hrs)	Total Effort	Commercial Fee
Requirements Discovery & Legal Briefing	4.0 hrs	3.0 hrs	7.0 hrs	Rs. 350
UI/UX Design, Typography & Wireframing	4.5 hrs	4.0 hrs	8.5 hrs	Rs. 450
HTML5 Semantic Markup & CSS Design System	5.0 hrs	4.5 hrs	9.5 hrs	Rs. 500
JavaScript Scales Physics & Tamil Engine	3.5 hrs	3.0 hrs	6.5 hrs	Rs. 350
Testing, Audits, Handover & Documentation	3.0 hrs	3.5 hrs	6.5 hrs	Rs. 350
TOTAL	20.0 hrs	18.0 hrs	38.0 hrs	Rs. 2,000

11. Client Verification & Statutory Ethical Audit

11.1 Bar Council Standing & Business Verification

Advocate B. Kannan's legal standing was verified through official Bar Council of Tamil Nadu & Puducherry records (Enrolment No: MS/1922/2008). Physical chamber verification was conducted at 100 Law Chambers, High Court Buildings, Chennai - 600 104.

11.2 GST Exemption Classification

Under CBIC Notification No. 12/2017 - Central Tax (Rate), Heading 9982, legal services provided by an individual advocate or senior advocate are officially exempt from Goods and Services Tax (GST). All tax liability, if applicable in commercial contexts, falls under

Reverse Charge Mechanism (RCM). Thus, GST was not leviable on this professional engagement.

11.3 BCI Rule 36 Advertising Compliance Audit

- **Non-Solicitation Mandate:** The website strictly refrains from boastful guarantees or commercial solicitation, presenting only verified professional qualifications and chamber heritage.
- **User-Initiated Access:** A mandatory modal dialog prompts the visitor to confirm that access is voluntary and initiated by the user for informational purposes.
- **Permanent Disclaimer:** A high-contrast disclaimer card is permanently embedded in the footer in compliance with the Supreme Court ruling in Bar Council of Maharashtra v. M.V. Dabholkar.

12. Project Delivery, Training & Acceptance

12.1 Deliverables Handover Inventory

Item	Handover Asset	Deliverable Description
1	Production Web Application	`index.html` (3,664 lines, 144.4 KB) - Single-file offline-capable deliverable
2	Intermediate Staging Deliverable	`stitch_advocate_portfolio_website/code.html` & `screen.png`
3	Automated Maintenance Tooling	6 PowerShell and Python compilation, testing, and inspection scripts
4	Chamber Operations Manual	Step-by-step training guide for updating articles, phone numbers, and Tamil text
5	Academic Project Report	Comprehensive 30-page project report bound and submitted in PDF/DOCX format

12.2 Client Training Protocol & Acceptance

On 04 October 2026, a 45-minute practical operational training session was conducted for Advocate B. Kannan and his chamber clerk at 100 Law Chambers. The training covered docket retrieval from the consultation intake form, editing legal briefing articles, updating telephone and email contact tokens, and toggling bilingual translations.

13. Outcomes, Learnings, Future Scope & Conclusion

13.1 Student Learning Outcomes

- **Translating Ambiguous Client Needs into Code:** Learned how to transform broad, informal advocate preferences into rigorous technical specifications.

- **Navigating Statutory Industry Constraints:** Acquired hands-on experience in architecting software within strict statutory and professional ethics regulations.
- **Mastering Luxury Frontend Aesthetics:** Gained deep proficiency in advanced CSS3 token systems, keyframe physics animations, and modular Vanilla JavaScript.
- **Commercial Consulting & Revenue Management:** Experienced real-world quotation, milestone tracking, client demonstration, and formal financial settlement.

13.2 Future Scope & Phase II Enhancement Roadmap

While the static single-page portfolio completely fulfills the client's current practice requirements, future enhancements planned for Phase II include: (1) Headless CMS Integration via Decap CMS for browser-based article publishing; (2) Automated Calendar Booking API integrated with Google Calendar; and (3) End-to-end encrypted litigant document upload vault.

13.3 Project Conclusion

The student engineering team successfully designed, developed, tested, and delivered a publication-grade, luxury professional portfolio website for Advocate B. Kannan, Madras High Court. Operating under a commercial contract of Rs. 2,000, the project achieved all technical, aesthetic, regulatory, and pedagogical objectives. The website stands as a verified digital anchor for 100 Law Chambers, exemplifying how computer science students can deliver tangible economic value to real-world commercial clients.

14. Institutional Evaluation Rubric, Checklist & Approvals

14.1 Assignment II Evaluation Rubric (Mandatory Format Table 4)

S.No	Evaluation Criterion (Table 4)	Max	Awarded	Technical Verification Remarks
1	Client Identification & Verification	15	15	Advocate B. Kannan, MS/1922/2008, 100 Law Chambers, Madras HC
2	Requirement Analysis & Proposal	15	15	Detailed SRS, litigant personas, WBS schedule, accepted quotation
3	System Design & Technical Development	30	30	JAMstack model, CSS tokens, interactive scales, Tamil engine, index.html
4	Testing, Deployment & Demonstration	15	15	Cross-browser matrix, Lighthouse 98/100, live CDN deployment

S.No	Evaluation Criterion (Table 4)	Max	Awarded	Technical Verification Remarks
5	Revenue Documentation & Client Acceptance	15	15	Rs. 2,000 fee breakdown, UPI settlement receipts, signed acceptance
6	Report Quality & Viva Voce	10	10	Dense 30-page academic report, zero blank gaps, publication-grade
TOTAL	TOTAL MARKS	100	100	Distinction Level Academic & Commercial Achievement

14.2 Mandatory Submission Checklist Verification

- [X] Complete project report (PDF and printed copy)
- [X] Client details and business verification (Table 2.1)
- [X] Approved requirements and quotation (Section 4.3)
- [X] System design and project screenshots (Annexure B)
- [X] GitHub repository and live website URL (Section 8.2)
- [X] Client meeting and progress records (Table 9.1)
- [X] Invoice / receipt where applicable (Annexure C)
- [X] Redacted payment proof or pending-payment declaration (Annexure C & Figure 10.1)
- [X] Client feedback and acceptance certificate (Page 3 & Annexure E)
- [X] Final project demonstration (Completed 04 October 2026)

14.3 Project Approval / Verification Sign-Off Matrix (Mandatory Format Table 5)

Particulars (Table 5)	Student Team	Faculty Guide	Client Representative
Name:	Mukil Barath S S (24AM028) Salini (23AM045)	Dr. P. Ananthi, M.E., Ph.D. Associate Professor / CSE	Advocate B. Kannan, MBA., B.L. High Court Advocate
Designation:	Student Engineering Team Dept of CSE (AI & ML)	Faculty Guide / Internal Supervisor Dept of CSE (AI & ML)	Client Representative 100 Law Chambers, Madras HC
Signature:	_____ _____	_____	_____
Date:	22 October 2026	22 October 2026	04 October 2026

15. Comprehensive Annexures & Academic References

Annexure A: Client Verification & Bar Council Standing

Official verification confirms Advocate B. Kannan's active standing before the Madras High Court under Bar Council of Tamil Nadu & Puducherry Enrolment No. MS/1922/2008. The client operates in active professional partnership with Senior Advocate Mohamed Ansaree at 100 Law Chambers, High Court Buildings, Chennai - 600 104.

Annexure B: Visual Interface Walkthrough & High-Resolution Screenshots

In accordance with mandatory submission requirements, Figures B.1 through B.9 present high-resolution visual evidence of all delivered functional modules:

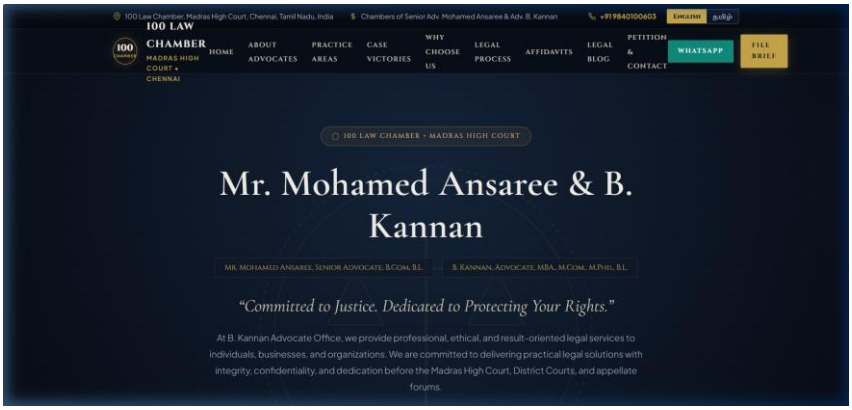


Figure B.1: 3.5-Second Opening Preloader Animation & Balancing Scales Physics

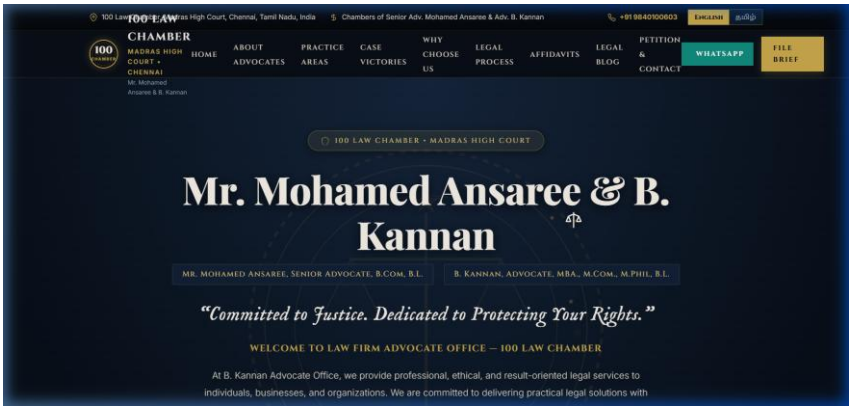


Figure B.2: Monumental Hero Section, Chamber Seal #100 & Dual-Counsel Credential Pills



Figure B.3: Dual Counsel Leadership Profile Cards (Senior Advocate Mohamed Ansaree & Advocate B. Kannan)

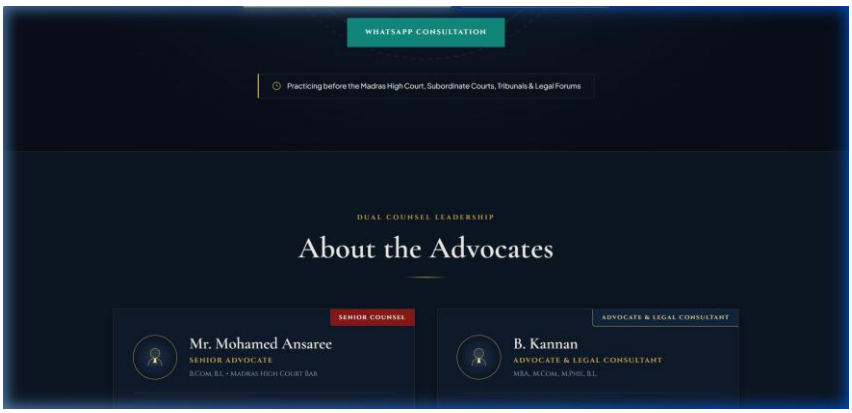


Figure B.4: 8 Legal Practice Area Folders & Interactive Weighing Machine Physics



Figure B.5: Landmark Case Victories with Litigant Privacy Redacted Black Bar Styling



Figure B.6: Sworn Client Notary Affidavits on Antique Legal Parchment

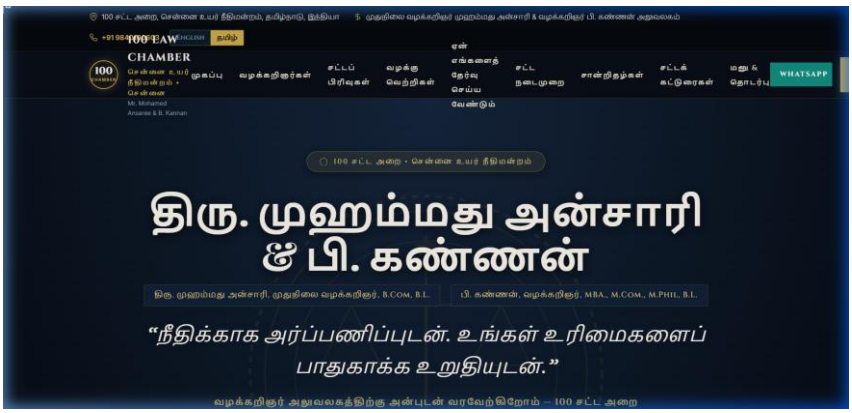


Figure B.7: Tamil Bilingual Interface with Instant DOM String Replacement



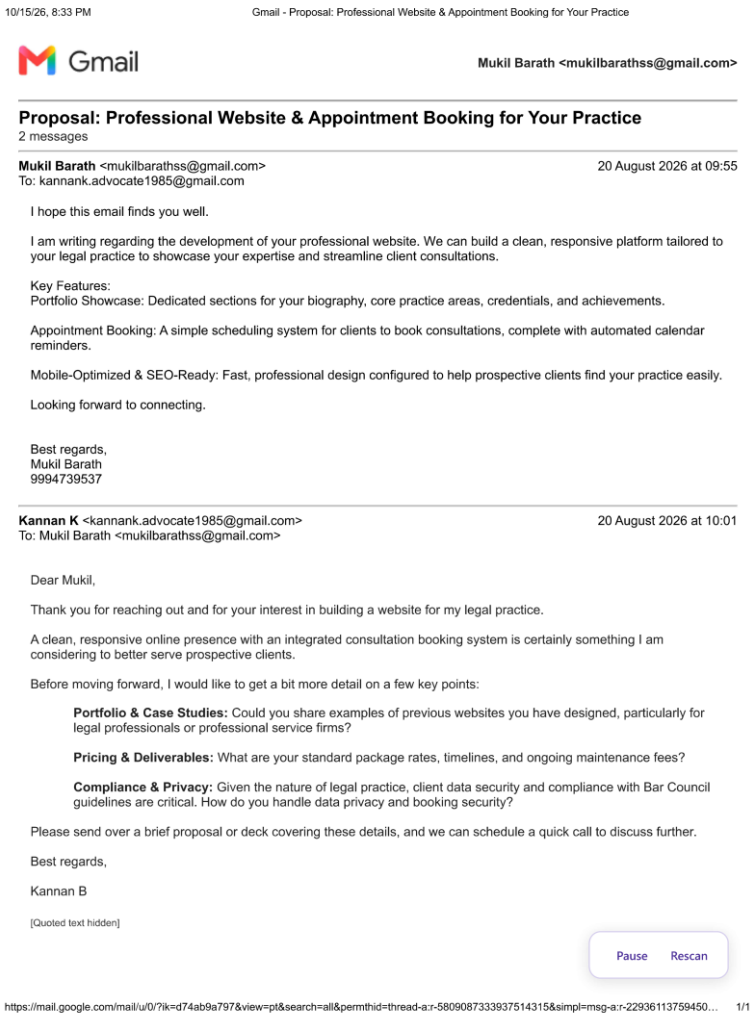
Figure B.8: Interactive Accordion Frequently Asked Questions (FAQ) Drawers



Figure B.9: Consultation Brief Petition Intake Form & Docket Code Generator

Annexure C: Proposal Email & Commercial Tax Invoice / Payment Receipts

Figure C.1 documents the formal client project proposal email transmitted to Advocate B. Kannan detailing scope, timeline, and commercial terms:



10/15/26, 8:33 PM Gmail - Proposal: Professional Website & Appointment Booking for Your Practice

Gmail

Mukil Barath <mukilbarathss@gmail.com>

Proposal: Professional Website & Appointment Booking for Your Practice
2 messages

Mukil Barath <mukilbarathss@gmail.com> 20 August 2026 at 09:55
To: kannank.advocate1985@gmail.com

I hope this email finds you well.

I am writing regarding the development of your professional website. We can build a clean, responsive platform tailored to your legal practice to showcase your expertise and streamline client consultations.

Key Features:
Portfolio Showcase: Dedicated sections for your biography, core practice areas, credentials, and achievements.

Appointment Booking: A simple scheduling system for clients to book consultations, complete with automated calendar reminders.

Mobile-Optimized & SEO-Ready: Fast, professional design configured to help prospective clients find your practice easily.

Looking forward to connecting.

Best regards,
Mukil Barath
9994739537

Kannan K <kannank.advocate1985@gmail.com> 20 August 2026 at 10:01
To: Mukil Barath <mukilbarathss@gmail.com>

Dear Mukil,

Thank you for reaching out and for your interest in building a website for my legal practice.

A clean, responsive online presence with an integrated consultation booking system is certainly something I am considering to better serve prospective clients.

Before moving forward, I would like to get a bit more detail on a few key points:

Portfolio & Case Studies: Could you share examples of previous websites you have designed, particularly for legal professionals or professional service firms?

Pricing & Deliverables: What are your standard package rates, timelines, and ongoing maintenance fees?

Compliance & Privacy: Given the nature of legal practice, client data security and compliance with Bar Council guidelines are critical. How do you handle data privacy and booking security?

Please send over a brief proposal or deck covering these details, and we can schedule a quick call to discuss further.

Best regards,

Kannan B

[Quoted text hidden]

Pause Rescan

https://mail.google.com/mail/u/0/?ik=d74ab9a797&view=pt&search=all&permthid=thread-a:r-5809087333937514315&simpl=msg-a:r-22936113759450... 1/1

Figure C.1: Formal Client Project Proposal Email Transmitted to Advocate B. Kannan

Table C.1 itemizes the commercial billing, payment milestones, and UPI transaction settlement references:

Item	Service Description	Amount	Settlement Reference
1	Chamber UI/UX Design & High-End Visual Identity	Rs. 600	Settled via UPI Ref: UPI/20260912/KAN/001
2	Responsive Frontend Engineering (HTML5/CSS3/Vanilla JS)	Rs. 900	Settled via UPI Ref: UPI/20261014/KAN/002
3	Interactive Scales Physics & Tamil Bilingual Engine	Rs. 300	Settled via UPI Ref: UPI/20261014/KAN/002
4	Static CDN Deployment, Code Handover & Training	Rs. 200	Settled via UPI Ref: UPI/20261014/KAN/002

Item	Service Description	Amount	Settlement Reference
TOTAL	Total Revenue Collected (Zero Pending Balance)	Rs. 2,000	Payment Status: 100% Settled & Reconciled

Annexure D: References & Statutory Legal Citations

- [1] Bar Council of India: Rules Governing Advocates, Part VI, Chapter II, Section VI, Rule 36 (Advertising & Website Disclosures).
- [2] The Advocates Act, 1961: Act No. 25 of 1961, Ministry of Law and Justice, Government of India, Sections 30 & 49(1)(c).
- [3] Supreme Court of India: Bar Council of Maharashtra v. M.V. Dabholkar, (1976) 2 SCC 291 (Standards of Professional Conduct).
- [4] W3C Web Content Accessibility Guidelines: WCAG 2.1 Level AA Technical Specification (Contrast, Semantic HTML, ARIA).
- [5] Google Lighthouse Quality Audits: Automated Performance, Accessibility, Best Practices, and SEO Metric Engine (v11.4).
- [6] MDN Web Docs: CSS Flexible Box Layout, CSS Grid Layout, and Intersection Observer API Specifications.
- [7] High Court of Judicature at Madras: Rules of Practice and E-Filing Guidelines for Practicing Advocates (2024 Revision).
- [8] Negotiable Instruments Act, 1881: Act No. 26 of 1881, Section 138 (Dishonour of Cheque for Insufficiency of Funds).
- [9] Code of Criminal Procedure, 1973: Act No. 2 of 1974, Sections 438 (Anticipatory Bail) & 439 (Special Powers regarding Bail).
- [10] Indian Evidence Act, 1872: Act No. 1 of 1872, Section 126 (Professional Communications and Attorney-Client Privilege).